THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 11: **Patterns in Life: Diversity and Classification** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Linnaeus's naming system is used worldwide. Reason (R): A scientific name consists of a genus and a species name in Latin form. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. An animal has radial symmetry, a spiny skin and a water vascular system. It is an: 1

(a) annelid **(b)** cnidarian **(c)** echinoderm **(d)** mollusc

3. Which group of organisms is the most diverse in the number of species? 1

(a) reptiles **(b)** mammals
(c) arthropods (insects) **(d)** birds

4. Organisms like Euglena show both plant-like and animal-like features. This indicates: 1

(a) Euglena is a plant **(b)** Euglena is a fungus
(c) Euglena is an animal **(d)** classification boundaries are not rigid

5. The kangaroo is a: 1

(a) reptile **(b)** marsupial mammal **(c)** bird **(d)** amphibian

6. Assertion (A): Birds are warm-blooded animals. Reason (R): They maintain a constant body temperature. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. An animal is aquatic, has gills, scales and fins and is cold-blooded. The class is: 1

(a) Amphibia **(b)** Reptilia **(c)** Pisces **(d)** Aves

8. The group of plants with the largest number of species is: 1

(a) angiosperms **(b)** pteridophytes **(c)** gymnosperms **(d)** bryophytes

9. **Assertion (A):** Mushrooms are not classified as plants. 1
Reason (R): Fungi lack chlorophyll and feed on dead organic matter (saprotrophic nutrition).
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Fish have a two-chambered heart. 1
Reason (R): Fish are cold-blooded animals.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are sponges considered animals even though they are sessile? 2

12. State why the animal kingdom is divided into vertebrates and invertebrates and which one is more diverse. 2

13. Why is a silkworm classified as an insect and a spider as an arachnid? 2

14. State two features that place a platypus in the class Mammalia though it lays eggs. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is the parasitic mode of life common in Platyhelminthes and Nematoda? State two adaptations. 3

16. How do you decide whether two organisms belong to the same species? Give the criteria with an example of the horse and the donkey. 3

17. Explain why the platyhelminthes are called acoelomate and the annelids coelomate. 3

OR

A student says 'a bat is a bird because it can fly'. Respond with three facts. 3

18. Describe the characteristics of the Class Insecta and give the names of three orders. 3

OR

Compare the classification by Linnaeus (two kingdoms) with that of Whittaker (five kingdoms) in terms of the placement of fungi and bacteria. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** During a nature walk in the Western Ghats, Divya collects a moss, a fern, a pine cone, a rose flower and a mushroom. She is asked to classify them. 5
(i) Name the group to which the moss belongs.
(ii) Which specimen bears seeds that are not enclosed in a fruit?
(iii) Which specimen is an angiosperm?
(iv) Why is the mushroom not placed with the plants?

20. Analyse the statement 'Classification is both a science and an art' with reference to the choice of characters, hierarchy, and the discovery of new species and molecular data. 5

OR

Explain how biodiversity hotspots are identified and the importance of the Western Ghats and the Himalayas, with measures to conserve their species. 5

END OF QUESTION PAPER **Total: 40 Marks**